



Seasonal Exploratory Analysis of NHS Appointment Data Using Python

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1. Background & Context

Using the two questions posed by the NHS regarding its appointment data, '**Has there been adequate staff and capacity in the networks?**' and '**What was the actual utilisation of resources?**', two business problem statements were developed to inform the following analysis of NHS appointment data:

- Investigate the correlation between appointment wait time and patient attendance rates across healthcare providers to determine if increased delays directly reduce patient attendance and recommend appropriate communication strategies.
- Evaluate the relationship between seasonality and staff to patient ratios to identify periods of misaligned staff supply and provide recommendations to optimise workforce scheduling.

2. Analytical Approach

NHS data was imported into Python alongside relevant libraries (Pandas, Numpy, etc). All datasets were scanned for null values. Initially, it seemed no missing values were present but upon inspection, missing data was tracked as 'Unknown' or 'Unmapped'. A variable was developed containing all of these missing values and used to replace those values as a 'null value' so as not to skew the data. This was repeated on all datasets.

```
# Creating a variable for missing values
missing_variants = ['Unknown', 'Unknown / Data Quality', 'Inconsistent Mapping', 'Unmapped']

# Replacing unknown datapoints with null values
ad_clean = ad.replace(missing_variants, np.nan)

# Determining sum of null values per column
ad_clean.isnull().sum()
```

Figure 2.1

The NC (National Categories) data and the AR (Appointments Regional) data were partially merged to link the 'sub_icb_location_name' in NC with the 'icb_location_code' in AR. A full merge was not useful in this instance as the datasets covered similar date ranges and, without unique appointment IDs, risked duplicating information. Linking dataframes through ICB codes strengthened the locational context of the AR dataframe.

Additionally, using the data within the AR DataFrame, a column was calculated representing a location's 'Did Not Attend' rate based on the following calculation: ('DNA_appointments' / 'count_of appointments'). See figure 2.2 below

	icb_ons_code	appointment_month	hcp_type	Total_Appts	DNA_Appts	DNA_Rate
1159	E54000036	2020-10	Other Practice staff	60	31	51.666667
2073	E54000055	2021-05	Other Practice staff	76	39	51.315789
2474	E54000062	2020-08	GP	59	30	50.847458
2462	E54000062	2020-02	GP	63	32	50.793651
2519	E54000062	2022-06	Other Practice staff	62	31	50.000000
...	...	...	...	...	...	...
2126	E54000056	2021-02	GP	55	19	34.545455
1416	E54000040	2021-07	GP	47	16	34.042553
1590	E54000043	2021-04	GP	56	19	33.928571
2124	E54000056	2021-01	GP	56	19	33.928571
1386	E54000040	2020-04	GP	43	14	32.558140

Figure 2.2

The average wait time per location was also calculated based on the 'time_between_booking_appointment' column and the 'count_of_appointments'. Data was weighted into tiers based on the time. For example, if the value for 'time_between_book_and_appointment' was '22 to 28 Days' that row was assigned a weighted value of 25. The weighted value was then multiplied by the count of appointments to assign a total weighted days value per row. The data was then aggregated on 'sub_icb_location_name', whilst the totals for each location were identified by dividing the total weighted days by the count of appointments.

```
# creating a new dataframe to work off of to perform analysis
ar_wait_times = ar_clean

# Making sure all values have the same syntax
ar_wait_times['time_between_book_and_appointment'] = ar_wait_times['time_between_book_and_appointment'].str.strip()

# Creating a dictionary to change wait times to numeric values
wait_mapping = {
    'Same Day' : 0,
    '1 Day' : 1,
    '2 to 7 Days' : 4.5,
    '8 to 14 Days' : 11,
    '15 to 21 Days' : 18,
    '22 to 28 Days' : 25,
    'More than 28 Days' : 35
}

# Add a new column to existing df and apply the wait mapping to the time between appts column
ar_wait_times['wait_numeric'] = ar_wait_times['time_between_book_and_appointment'].map(wait_mapping)

# Calculating the amount of days that were weighted
ar_wait_times['total_weighted_days'] = ar_wait_times['wait_numeric'] * ar_wait_times['count_of_appointments']

# View the data
ar_wait_times
```

Figure 2.3

Further analysis on the breakdown of GPs versus Other Practitioners, led to a clear finding in the differences in waiting times. By aggregating this way, we were able to analyse the locations with the lowest and highest wait times in order to compare their staffing split and identify possible trends.

GP appointments, attendance rates and appointment wait times were analysed seasonally and split by HCP type (Healthcare Provider type). Utilisation was also aggregated seasonally against the 2019 benchmark of 24.2 million appointments a month. By aggregating seasonally, the analysis aimed to provide a clear breakdown of trends whilst identifying months with commonalities in utilisation values or appointment data. In order to divide each dataframe seasonally, they were filtered based on specific monthly breakdowns. Seasonal aggregations syntax can be seen below:

```
# Creating seasonal filters
summer = nc_ss_day[nc_ss_day['appointment_month'].isin(['2021-06', '2021-07', '2021-08'])]
autumn = nc_ss_day[nc_ss_day['appointment_month'].isin(['2021-09', '2021-10', '2021-11'])]
winter = nc_ss_day[nc_ss_day['appointment_month'].isin(['2021-12', '2022-01', '2022-02'])]
spring = nc_ss_day[nc_ss_day['appointment_month'].isin(['2022-03', '2022-04', '2022-05'])]
```

Figure 2.4

NHS provided Twitter data was also analysed for trends and public sentiments. All hashtags from the dataset were extracted, stripped for consistency and sorted by frequency. The resulting list showed that general terms like 'Healthcare', 'health' and 'medicine' were among the most popular. Data was filtered for mentions of the nhs using .str.contains() where only three tweets mentioned the NHS, none of which pertained to patient/staff experience or resource utilisation.

Further analysis filtered for country specific keywords to determine if the data collected was representative of the UK or a broader international context. There were over 80 tweets relating to American and Canadian keywords, but only 6 for the UK centric words. The presence of 'medicare' in our top hashtags list also implies a US centric dataset. Based on these findings, the data quality has been deemed unfit to guide resource or staffing recommendations. It would be useful to collect geographically relevant twitter data in order to accurately capture geographically relevant public sentiment.

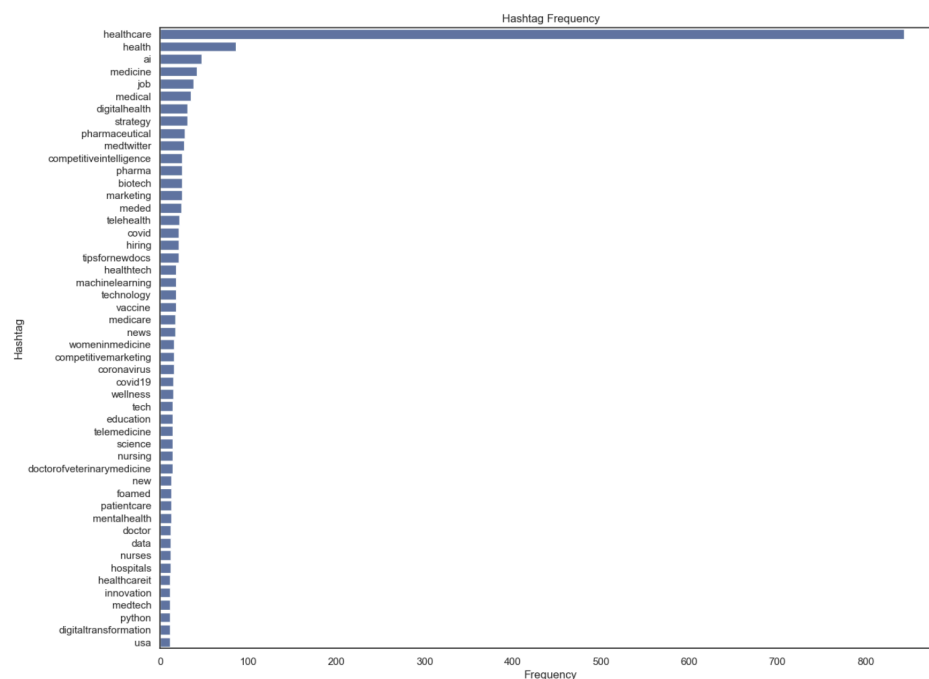


Figure 2.5

3. Visualisations and Insights

To guide our recommendations, three key visualizations will be explored to address our business problem statements. Firstly, we have a breakdown of the trends in appointment volume per HCP type alongside a benchmark line representing the overall average monthly appointments per HCP per month. A bar chart was chosen to capture monthly totals and visualise specific peaks and troughs based on our benchmark. Here, we see that Autumn surpasses this benchmark for both HCP types and underperforms in winter for both HCP types.

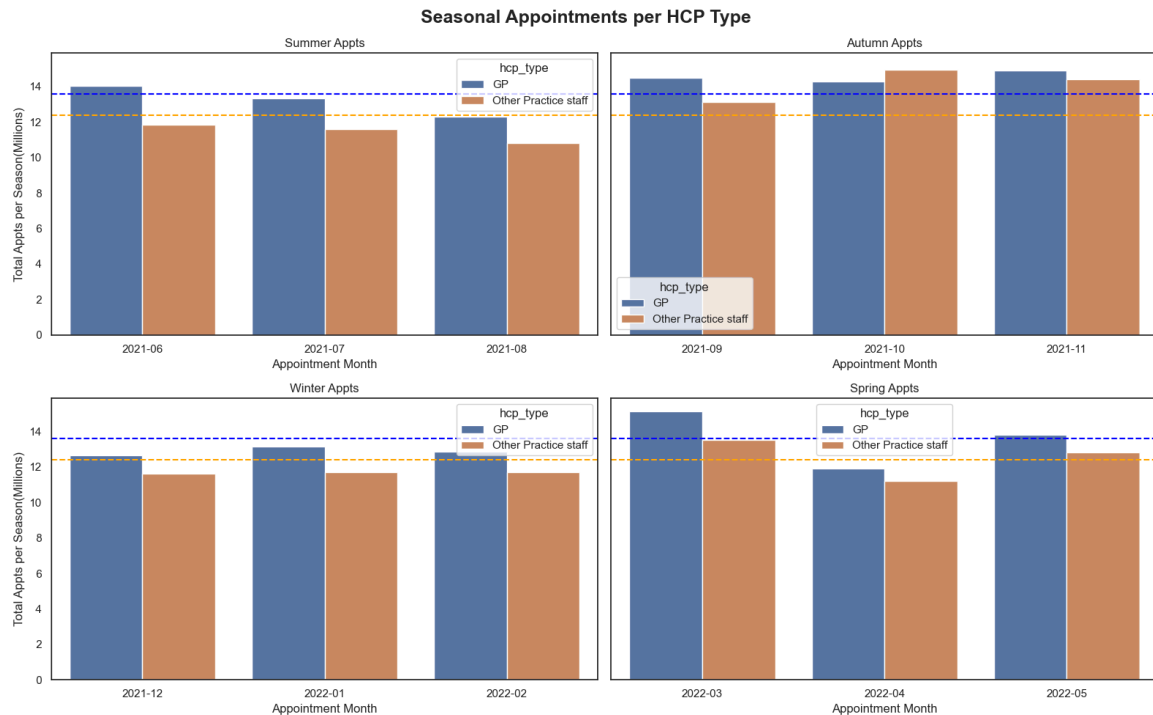


Figure 3.1

Building on these findings, the second visualisation analyses the time between booking an appointment and attending the appointment. Again, a bar chart form is chosen alongside general average benchmark lines to provide a base analysis level. This visualisation clearly shows Autumn witnesses a much higher than average time than other months in terms of waiting time for both HCP types.

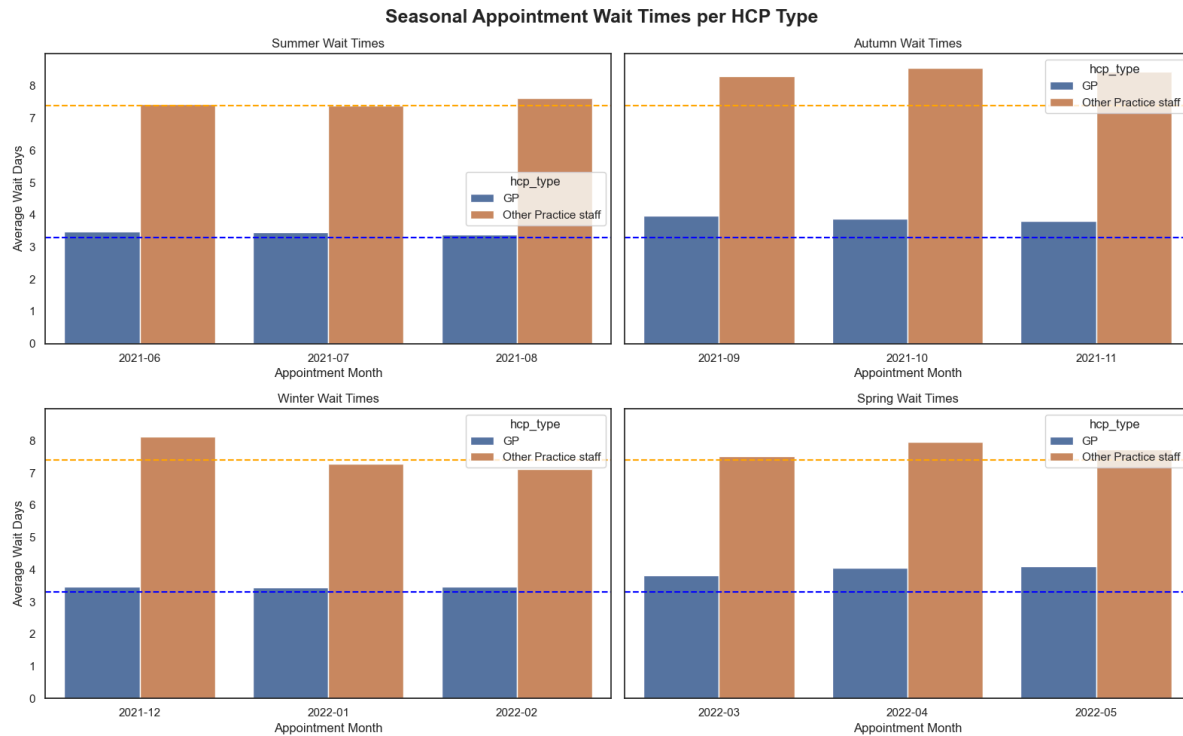


Figure 3.2

The third visualisation analyses overall NHS utilisation based on the 2019 average of 24.2 million appointments per month. We can see that the NHS actually surpasses the baseline utilisation of 1 (reaching 24.2 million appointments per month) in all but 2 months of the year (August and April). This means that the NHS is surpassing their standard level of operating capacity the majority of the time.

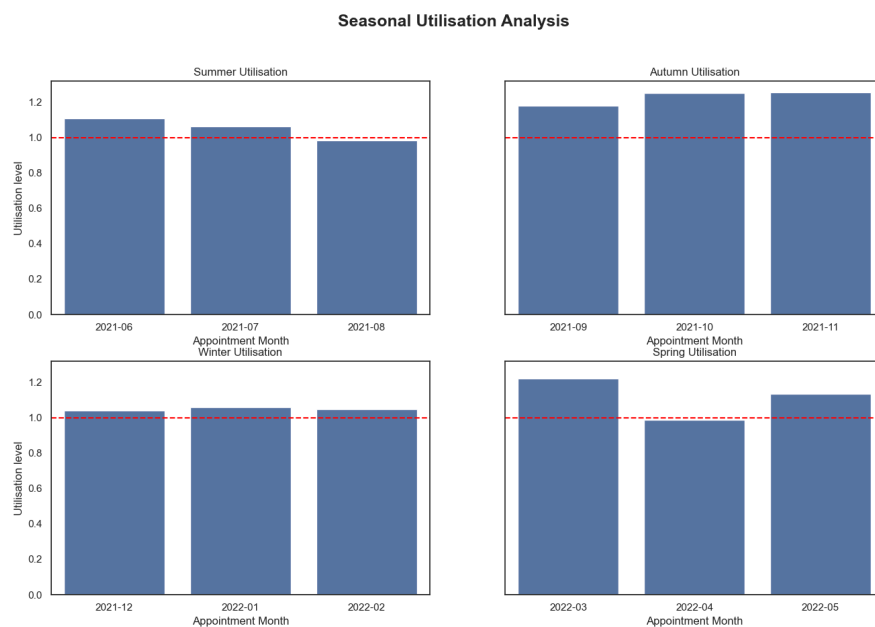


Figure 3.3

Overall, the actual utilisation of staff resources is being met, but there is clear evidence suggesting that there is demand for more resources in specific months - most specifically in Autumn and the beginning and end of spring (March & May). We see increased wait times, an increased number of appointments being scheduled and an over utilisation of staff resources. This may be in alignment with the return back to school for children where flu cases are twice as high as any other demographic ([NHS, 2024](#)). The potential influx of flu related appointments may be taking up more staffing resources than steadily available.

4. Patterns and Predictions

Based on seasonal trends and high intensity utilisation periods identified through our visualisations, the following recommendations are made to address the questions posed by the NHS:

Has there been adequate staff capacity in the networks?

No. based on the increased volume of appointments, wait time and utilisation levels, we can conclude that staff may be overrun specifically in the Autumn months. As sheer appointment volume and length of appointment wait time peak in Autumn, we can infer that these factors are correlated and because of the increase in demand (scheduled appointments) the supply (GP and OP availability) does not match thus driving up wait times. We recommend hiring more staff during these months to match the increase in demand.

What was the actual utilisation of resources?

The utilisation analysis uncovered that the NHS is consistently operating at a high appointment saturation level where actual utilisation frequency exceeds the 2019 baseline of 24.2 million appointments per month. Our utilisation visualisation clearly shows that the NHS is operating above the utilization index of 1 in 10 out of 12 months of the year. Coinciding with our previously determined seasonal peak times, utilisation reaches up to 125% of the standard operating capacity in autumn. Because utilisation exceeds 100% so frequently, we can conclude that staff are operating at an unsustainably high rate.

Recommendations:

- **Seasonal staffing:** Implement increased seasonal staffing specifically in Autumn (September-November) to address the 25% increase in operational utilisation during his period.
- **Data Collection Improvements:** Future analysis would benefit from refined social media (Twitter) scraping with regionally specific filters to allow for accurate UK based sentiment analysis on patient and staffing trends.
- **Communication Strategy:** Develop targeted text reminders for patients with appointment wait times of over 22+ days to help minimise DNA rates and avoid wasted appointment slots

Appendix

2.1 Missing variables

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# Creating a variable for missing values
missing_variants = ['Unknown', 'Unknown / Data Quality', 'Inconsistent Mapping', 'Unmapped']

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2.2 DNA Rates DataFrame

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2.3 Wait Time mapping

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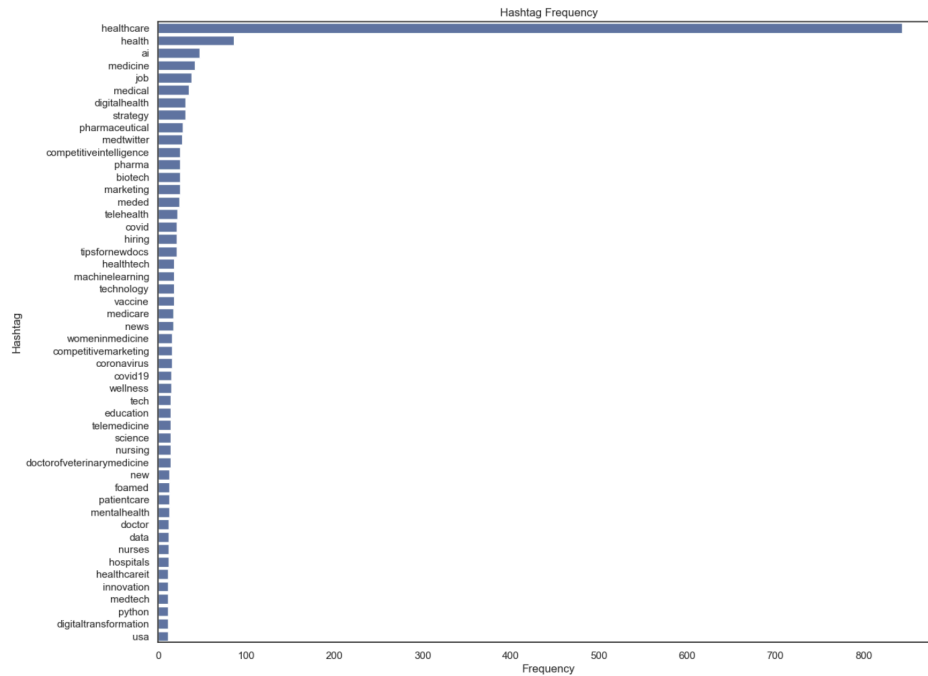
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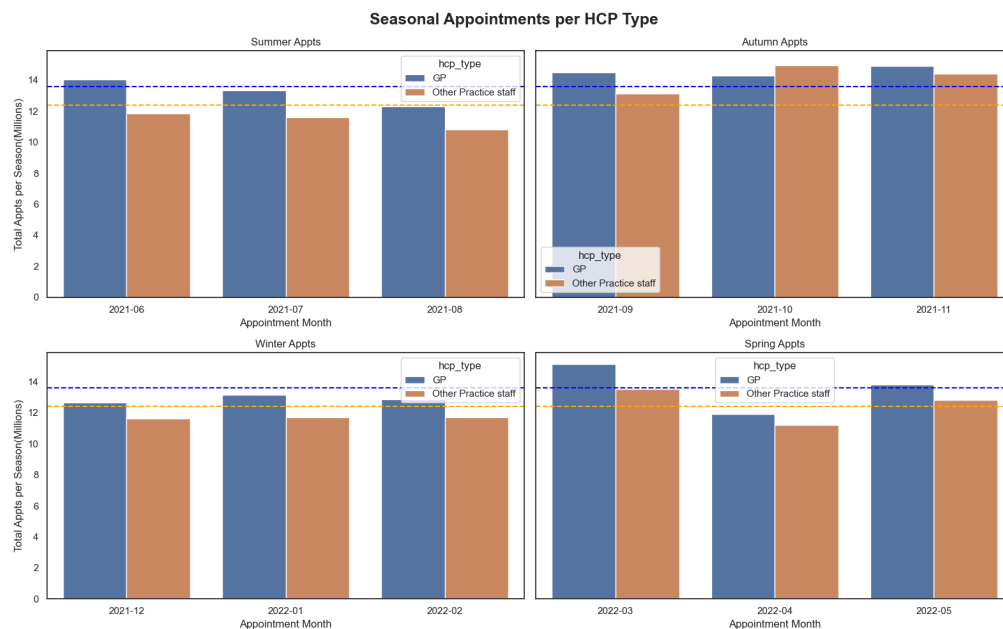
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```

2.5 Hashtag Frequency



3.1 Seasonal Appointments



3.3 Seasonal Utilisation analysis

